Week 4 question sheet

Problem sheet

Instructions

Attempt all ten questions and all 17 scorable parts. The target time is 240 minutes and the total is 100 marks. No calculator or software is needed. Give exact values. Write gradients and displacement vectors as columns, put one output component in each Jacobian row, and state dimensions before using the multivariable chain rule.

Questions 9 and 10 form the earlier-outcome retrieval block. Complete them without consulting Weeks 1–3 notes. Use progressive hints only after a serious attempt, and do not open the complete solutions until you have attempted or paused every part.

Question 1 — partial derivatives and a Hessian

Let

$$f(x, y) = x^2 e^y + y \log x, \qquad (x, y) \in (0, \infty) \times \mathbb{R}.$$

Part A — `W04-Q01-PA` (5 marks; 10 minutes) Calculate $f_x(x, y)$ and $f_y(x, y)$, write the column gradient, and evaluate it at $(1, 0)$.

Part B — `W04-Q01-PB` (6 marks; 12 minutes) Calculate all four second partial derivatives and $\mathbf{H}_f(1, 0)$. State why the off-diagonal entries agree on the stated domain.

Question 2 — a gradient and local change

Define

$$g(x, y, z) = xy^2 + e^{z-x}, \qquad (x, y, z) \in \mathbb{R}^3.$$

Part A — `W04-Q02-PA` (6 marks; 12 minutes) Calculate the column gradient $\nabla g(x, y, z)$ and evaluate it at $\mathbf{a} = [1, 2, 1]^\mathsf{T}$.

Part B — `W04-Q02-PB` (6 marks; 14 minutes) Use the first-order differential to approximate the change in g from $\mathbf{a}$ under the displacement

$$\mathrm{d}\mathbf{x} = \begin{bmatrix} 0.02 \\ -0.01 \\ 0.03 \end{bmatrix}.$$

Show the dimensions of the product. Explain why the result is a scalar and why it is an approximation rather than an exact finite change.

Question 3 — a directional second derivative

Part A — `W04-Q03-PA` (7 marks; 16 minutes) Let

$$f(x, y) = x^3 + 2xy^2 - 6xy.$$

Calculate $\mathbf{H}_f(1, -1)$. For $\mathbf{v} = [1, 2]^\mathsf{T}$, calculate $\mathbf{v}^\mathsf{T}\mathbf{H}_f(1, -1)\mathbf{v}$. Then define

$$\varphi(t) := f(1+t, -1+2t)$$

and verify by direct one-variable differentiation that $\varphi''(0)$ equals your quadratic expression. Do not make a curvature or optimisation classification.

Question 4 — scalar chain rules

Part A — `W04-Q04-PA` (5 marks; 12 minutes) Let

$$f(x,y) = x^2 y, \qquad \mathbf{g}(t) = \begin{bmatrix} 1+t \\ e^t \end{bmatrix}.$$

Use the column-gradient form of the chain rule to calculate

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{g}(t))\Big|_{t=0}.$$

Show the dimensions of the gradient and path derivative.

Part B — `W04-Q04-PB` (7 marks; 16 minutes) Let

$$F(u,v) = uv + \log v, \qquad v > 0,$$

and

$$\mathbf{G}(x,y) = \begin{bmatrix} x^2 + y \\ e^y \end{bmatrix}.$$

For $h := F \circ \mathbf{G}$, calculate the column gradient $\nabla h(1,0)$ using the multivariable chain rule. State the dimensions of every factor. Verify the result by first writing $h(x,y)$ explicitly and differentiating it directly.

Question 5 — a vector-valued derivative

Define

$$\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^2, \qquad \mathbf{F}(x,y,z) = \begin{bmatrix} xy + z^2 \\ e^x + yz \end{bmatrix}.$$

Part A — `W04-Q05-PA` (6 marks; 12 minutes) Calculate $\mathbf{J}_{\mathbf{F}}(x,y,z)$, state its dimensions, and evaluate it at $\mathbf{a} = [0,1,2]^{\mathsf{T}}$.

Part B — `W04-Q05-PB` (7 marks; 16 minutes) For

$$\mathrm{d}\mathbf{x} = \begin{bmatrix} 0.02 \\ -0.01 \\ 0.01 \end{bmatrix},$$

calculate the first-order output change $\mathrm{d}\mathbf{F} = \mathbf{J}_{\mathbf{F}}(\mathbf{a})\mathrm{d}\mathbf{x}$. State the dimension of the result and interpret each of its two entries.

Question 6 — composition of vector-valued maps

Let

$$\mathbf{g} : \mathbb{R}^2 \to \mathbb{R}^3, \qquad \mathbf{g}(s,t) = \begin{bmatrix} s+t \\ st \\ s^2 \end{bmatrix},$$

and

$$\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^2, \qquad \mathbf{F}(x,y,z) = \begin{bmatrix} x+y+z \\ xy \end{bmatrix}.$$

Write $\mathbf{H} := \mathbf{F} \circ \mathbf{g}$.

Part A — `W04-Q06-PA` (7 marks; 16 minutes) Calculate $\mathbf{J}_{\mathbf{g}}(1,2)$ and $\mathbf{J}_{\mathbf{F}}(\mathbf{g}(1,2))$. State the dimensions of the two matrices and use the chain rule to find $\mathbf{J}_{\mathbf{H}}(1,2)$.

Part B — `W04-Q06-PB` (8 marks; 18 minutes) Write both components of $\mathbf{H}(s,t)$ explicitly and differentiate them directly to verify your answer to Part A. Explain why reversing the two Jacobian factors cannot give the derivative of $\mathbf{H} : \mathbb{R}^2 \to \mathbb{R}^2$, even though the reversed matrix product happens to be defined.

Question 7 — marginal effects in a production index

Part A — `W04-Q07-PA` (8 marks; 18 minutes) Consider

$$Y(K,L) = 2K^{1/2}L^{1/2}, \qquad K > 0, \quad L > 0.$$

At $(K,L) = (4,9)$:

1. calculate the column gradient and Hessian;
2. use the first-order differential to approximate the change in Y when $\mathrm{d}K = 1/10$ and $\mathrm{d}L = -3/20$; and
3. interpret the two first partial derivatives and explain the limitation of the finite-change approximation.

Do not classify the curvature of Y or solve an optimisation problem.

Question 8 — a Hessian through a scalar composition

Part A — `W04-Q08-PA` (10 marks; 24 minutes) Suppose

$$h(\mathbf{x}) = \phi(g(\mathbf{x})), \qquad g : \mathbb{R}^2 \to I, \qquad \phi : I \to \mathbb{R},$$

where $I \subseteq \mathbb{R}$ is an open interval containing the relevant values of g, and both functions have continuous second derivatives near the relevant points.

1. Starting from $\nabla h = \phi'(g)\nabla g$, derive

$$\mathbf{H}_h = \phi''(g)\nabla g \nabla g^{\mathsf{T}} + \phi'(g)\mathbf{H}_g.$$

   State the dimension of each term.

2. Apply the formula to

$$h(x,y) = \log(1 + x^2 + 2y^2)$$

with $I = (-1, \infty)$, and calculate $\mathbf{H}_h(1, 1)$.

3. Check one off-diagonal entry by differentiating h directly.

Do not use the Hessian to make a convexity or concavity classification.

Question 9 — Mathematics retrieval

Complete both parts without consulting earlier notes.

Part A — `W04-Q09-PA` (3 marks; 10 minutes; `M3` retrieval) Construct the second-order Taylor approximation to e^x about $x = 0$ and use it to approximate $e^{0.1}$. State that the resulting number is an approximation.

Part B — `W04-Q09-PB` (3 marks; 10 minutes; `M4` retrieval) Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}.$$

State the dimensions of $\mathbf{A}$, $\mathbf{b}$ and $\mathbf{Ab}$, then calculate the product.

Question 10 — Statistics consolidation

Complete both parts without consulting Week 3 notes. These are retrieval parts, not new Week 4 Statistics outcomes.

Part A — `W04-Q10-PA` (3 marks; 12 minutes; `S3` retrieval) Let

$$X \sim \text{Exponential}(2),$$

where 2 is the rate. State the support and calculate exactly

$$\mathbb{P}\left(\frac{1}{2} < X \leq 1\right).$$

Part B — `W04-Q10-PB` (3 marks; 12 minutes; `S4` retrieval) Suppose

$$\mathbb{E}[X] = 1, \qquad \mathbb{E}[Y] = 2, \qquad \text{Var}(X) = 2, \qquad \text{Var}(Y) = 3, \qquad \text{Cov}(X, Y) = 1.$$

For $Z = 2X - Y$, calculate $\mathbb{E}[Z]$ and $\text{Var}(Z)$. Explain the sign of the covariance term in the variance calculation.

Self-test questions

Set aside 24 minutes and attempt all four parts without the reading guide, worked examples, hints or exercise solutions. No calculator is needed. Show the dimensions or condition that justifies each method. The self-test has 20 marks and does not contribute to Checkpoint A or either mock-examination score.

Questions

`W04-ST01-PA` — a gradient and Hessian (`M6`; 5 marks; 6 minutes) Let

$$f(x, y) = x^2 + xy + y^2.$$

Calculate the column gradient at $(1, -1)$ and the Hessian at the same point.

W04-ST02-PA — a Jacobian and its dimensions (M6; 5 marks; 6 minutes) Define

$$\mathbf{F} : \mathbb{R}^2 \to \mathbb{R}^3, \qquad \mathbf{F}(x, y) = \begin{bmatrix} x + y \\ xy \\ x^2 \end{bmatrix}.$$

State the dimensions of the Jacobian and calculate $\mathbf{J}_{\mathbf{F}}(1, -1)$.

W04-ST03-PA — a scalar-path chain rule (M6; 5 marks; 6 minutes) Let

$$F(x, y) = xy + \log x, \qquad x > 0,$$

and

$$\mathbf{g}(t) = \begin{bmatrix} 1 + t \\ 1 - 2t \end{bmatrix}.$$

Use the column-gradient chain rule to calculate

$$\frac{\mathrm{d}}{\mathrm{d}t} F(\mathbf{g}(t))\Big|_{t=0}.$$

Show the dimensions of the product.

W04-ST04-PA — Statistics consolidation (S4; 5 marks; 6 minutes) Suppose $\mathbb{E}[X] = 3$ and $\mathrm{Var}(X) = 2$. For $Y = 1 - 2X$, calculate $\mathbb{E}[Y]$ and $\mathrm{Var}(Y)$. Explain why adding 1 affects the expectation but not the variance.